

STUTTGART, AR, USA

WMO: 723416

Lat: 34.600N

Lon: 91.574W

Elev: 224

StdP: 14.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	25.1	7.3	8.1	24.7	11.8	10.1	28.2	26.8	46.7	24.9	46.0	10.9	20	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	97.2	77.9	94.7	77.6	92.6	76.9	80.7	92.0	79.7	91.5	78.6	89.6	8.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	144.0	86.9	76.2	138.0	85.7	75.2	133.1	84.7	44.4	93.1	43.2	91.8	42.1	90.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.8	18.8		16.0	100.3	4.6	3.6	12.7	102.9	10.1	105.0	7.5	107.1	4.2	109.7
				13.1	82.3	4.2	1.6	10.1	83.5	7.6	84.4	5.3	85.3	2.3	86.4
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	41.8	46.0	53.7	63.2	72.2	80.0	82.3	81.4	74.6	63.5	52.4	44.7
	DBStd	16.39	10.18	10.23	9.70	7.94	6.81	4.65	4.04	4.39	6.31	8.40	8.70	9.28
	HDD50	845	292	181	72	6	0	0	0	0	0	4	72	218
	HDD65	2925	720	535	366	126	22	0	0	0	8	133	386	629
	CDD50	5613	38	68	188	401	687	899	1001	972	737	423	144	55
	CDD65	2218	1	3	17	71	244	449	536	507	294	87	8	1
	CDH74	22555	1	9	94	515	2017	4756	6126	5513	2686	788	48	2
	CDH80	9950	0	0	8	97	699	2176	2946	2668	1110	241	5	0
Wind	WSAvg	8.8	10.1	10.4	10.7	10.6	9.3	7.8	6.8	6.3	7.2	8.0	9.2	9.5
Precipitation	PrecAvg	48.3	3.9	3.8	4.8	5.1	4.7	3.2	3.5	2.6	3.1	4.5	4.1	4.8
	PrecMax	78.5	7.2	8.3	11.1	13.0	12.6	5.7	8.7	6.2	6.7	14.1	10.2	10.4
	PrecMin	35.7	0.3	1.3	1.7	1.5	0.8	1.3	0.7	0.6	0.1	0.0	0.9	0.6
	PrecStd	10.4	1.9	1.7	2.3	3.0	2.3	1.5	2.1	1.4	1.8	3.3	2.2	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.1	74.9	81.3	86.1	92.5	97.4	99.3	99.9	96.6	90.2	80.6	72.4
		MCWB	63.9	65.0	66.2	69.8	73.3	76.5	78.6	78.7	76.3	71.8	68.3	65.3
	2%	DB	66.0	70.3	76.7	82.3	90.0	94.9	97.0	97.2	92.5	86.0	73.4	66.5
		MCWB	60.9	62.5	63.7	67.6	72.6	76.8	78.5	78.5	75.5	71.0	63.0	61.5
	5%	DB	62.9	66.0	72.6	80.7	87.7	92.6	93.4	93.5	90.1	82.1	70.4	63.4
		MCWB	57.4	59.6	61.7	66.6	72.0	76.0	78.1	77.9	74.9	67.5	62.4	58.4
	10%	DB	57.3	62.8	69.5	76.8	83.9	90.4	91.4	91.2	87.6	78.9	66.1	59.2
		MCWB	51.8	56.0	59.9	64.9	71.1	75.4	77.6	77.2	73.5	66.2	60.0	54.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	67.1	69.0	73.0	77.0	80.5	82.2	82.3	79.2	76.0	70.4	66.5
		MCDB	70.1	72.1	77.8	82.8	86.8	91.9	92.2	93.5	91.7	85.7	76.8	71.1
	2%	WB	62.0	64.0	66.0	70.5	75.3	79.0	80.8	80.9	77.5	73.7	66.4	63.1
		MCDB	65.2	68.6	74.2	79.7	84.6	90.9	91.6	92.5	88.5	82.2	71.5	65.9
	5%	WB	58.4	60.7	63.4	68.7	74.1	77.8	79.7	79.7	76.3	71.1	63.8	59.2
		MCDB	62.3	65.3	69.9	77.2	83.3	88.8	91.3	91.7	86.3	77.7	68.8	62.9
	10%	WB	53.1	56.7	60.9	66.7	72.6	76.8	78.7	78.4	75.0	68.6	60.2	55.1
		MCDB	57.0	61.5	67.2	74.4	81.7	87.2	89.7	89.4	83.9	75.6	65.4	58.8
Mean Daily Temperature Range	5% DB	MDBR	15.3	16.5	17.8	19.2	18.0	17.6	17.5	18.5	20.1	21.4	17.8	15.1
		MCDBR	20.9	21.9	21.5	21.8	20.8	19.5	19.1	20.9	22.3	25.0	22.2	19.5
		MCWBR	16.3	16.2	13.4	11.2	8.4	7.4	7.5	7.8	9.0	11.7	14.2	15.4
	5% WB	MCDBR	18.3	19.8	19.2	18.8	17.8	17.6	17.9	19.0	19.0	20.0	18.8	17.5
		MCWBR	16.4	16.1	13.5	11.7	8.7	7.7	8.1	8.2	8.5	11.3	14.9	15.2
Clear-Sky Solar Irradiance	taub	0.323	0.332	0.362	0.405	0.436	0.466	0.495	0.486	0.419	0.360	0.336	0.320	
	taud	2.487	2.455	2.397	2.273	2.247	2.177	2.129	2.140	2.316	2.460	2.483	2.507	
	Ebn at Noon	280	290	289	279	271	262	253	253	266	276	273	275	
	Edh at Noon	27	31	36	43	44	47	49	48	38	30	27	25	
All-Sky Solar Radiation	RadAvg	792	953	1327	1676	1855	2085	2012	1851	1578	1225	878	700	
	RadStd	69	141	122	117	126	142	104	139	116	159	97	75	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page